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United States Patent	12397365
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Sigl; Dennis Roland et al.

Systems, methods, and apparatus to preheat welding wire

Abstract

An example welding torch includes: a first contact tip configured to conduct welding current to a consumable electrode; a second contact tip configured to conduct preheating current to the consumable electrode; a cooling assembly configured to transfer heat from at least the first contact tip to coolant and to conduct the welding current through the cooling assembly; wherein the first contact tip and the cooling assembly are removable from the welding torch as a single unit.

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Appl. No.:	18/085127
Filed:	December 20, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230119789 A1	Apr. 20, 2023

Related U.S. Application Data

continuation parent-doc US 16005456 20180611 US 11590597 child-doc US 18085127
us-provisional-application US 62517537 20170609

Publication Classification

Int. Cl.: **B23K9/10** (20060101); **B23K9/14** (20060101); **B23K9/173** (20060101); **B23K9/235** (20060101); **B23K9/26** (20060101); **B23K9/29** (20060101); **B23K37/003** (20250101)

U.S. Cl.:

CPC **B23K9/1093** (20130101); **B23K9/1043** (20130101); **B23K9/14** (20130101); **B23K9/173** (20130101); **B23K9/235** (20130101); **B23K9/26** (20130101); **B23K9/295** (20130101); **B23K37/003** (20130101);

Field of Classification Search

CPC: B23K (37/003); B23K (9/1043); B23K (9/1093); B23K (9/14); B23K (9/173); B23K (9/235); B23K (9/26); B23K (9/295)

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Background/Summary

RELATED APPLICATIONS (1) This patent is a continuation of U.S. patent application Ser. No. 16/005,456, filed Jun. 11, 2018, entitled “Systems, Methods, and Apparatus to Preheat Welding Wire,” and claims priority to U.S. Provisional Patent Application Ser. No. 62/517,537, filed Jun. 9, 2017, entitled “Systems, Methods, and Apparatus to Preheat Welding Wire.” The entirety of U.S. Provisional Patent Application Ser. No. 62/517,537 is incorporated herein by reference.

BACKGROUND

- (1) This disclosure relates generally to welding and, more particularly, to welding torches and methods to provide wire preheating for welding.
 - (2) Welding is a process that has increasingly become ubiquitous in all industries. Welding is, at its core, simply a way of bonding two pieces of metal. A wide range of welding systems and welding control regimes have been implemented for various purposes. In continuous welding operations, metal inert gas (MIG) welding and submerged arc welding (SAW) techniques allow for formation of a continuing weld bead by feeding welding wire shielded by inert gas from a welding torch and/or by flux. Such wire feeding systems are available for other welding systems, such as tungsten inert gas (TIG) welding. Electrical power is applied to the welding wire and a circuit is completed through the workpiece to sustain a welding arc that melts the electrode wire and the workpiece to form the desired weld.
 - (3) While very effective in many applications, these welding techniques may experience different initial welding performance based upon whether the weld is started with the electrode “cold” or “hot.” In general, a cold electrode start may be considered a start in which the electrode tip and adjacent metals are at or relatively near the ambient temperature. Hot electrode starts, by contrast, are typically those in which the electrode tip and the adjacent metals' temperature are much more elevated, but below the melting point of the electrode wire. In some applications, it is believed that initiation of welding arcs and welds is facilitated when the electrode is hot. However, the current state of the art does not provide regimes designed to ensure that the electrode is heated prior to initiation of a welding operation.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example robotic welding system in which a robot is used to weld a workpiece using a welding tool, in accordance with aspects of this disclosure.
- (2) FIG. 2 illustrates an example liquid-cooled welding torch, in accordance with aspects of this disclosure.
- (3) FIG. 3 illustrates an exploded view of the example liquid-cooled welding torch of FIG. 2.
- (4) FIG. 4 is a more detailed depiction of an example liquid cooling assembly that may be used to implement the liquid cooling assemblies of FIG. 2.
- (5) FIG. 5 is a cross-section view of the liquid-cooled power cable assembly of FIG. 4.
- (6) FIG. 6 is an exploded view of the example liquid-cooled power cable assembly of FIG. 3.
- (7) FIGS. 7A, 7B, and 7C are views of the example cooler body of FIG. 6.
- (8) FIG. 8 is a cross-section view of the liquid cooling assembly and the quick-disconnect assembly of FIG. 2 coupled to the liquid cooling assembly of FIG. 4.
- (9) FIG. 9 illustrates the example welding assembly and the example liquid cooling assemblies of FIG. 2 disconnected from the remainder of the torch via the quick-disconnect assembly.
- (10) FIG. 10 is a cross-section view of the example welding assembly of FIG. 2.
- (11) FIGS. 11A, 11B, and 11C show an example implementation of the first contact tip of FIG. 10.
- (12) FIGS. 12A and 12B show an example implementation of the wire guide of FIG. 10.
- (13) FIGS. 13A, 13B, and 13C show an example implementation of the second contact tip of FIG. 10.
- (14) FIGS. 14A and 14B illustrate views of the example diffuser of FIGS. 3 and 10.
- (15) FIG. 15A illustrates a conventional robotic welding torch having a first tool center point distance and torch neck angle, in accordance with aspects of this disclosure.
- (16) FIG. 15B illustrates an example implementation of the liquid-cooled welding torch of FIG. 2 configured to replace the conventional torch of FIG. 15A while maintaining a same tool center

point distance and torch neck angle.

(17) FIG. 16 is a cross section of an example welding cable that may be used to provide cooling liquid, welding current, and preheating current to a welding torch, and to carry cooling liquid away from the welding torch, in accordance with aspects of this disclosure.

(18) FIG. 17 illustrates a functional diagram of an example welding system including the example welding torch of FIG. 2, and which may be used with the welding system of FIG. 1.

(19) FIG. 18 is a block diagram of an example implementation of the power supplies of FIG. 17.

(20) FIG. 19 illustrates another example liquid-cooled welding torch, in accordance with aspects of this disclosure.

(21) FIG. 20 is an exploded view of the example welding torch of FIG. 19.

(22) FIG. 21 is a cross-sectional plan view of the example welding assembly of FIG. 19.

(23) FIG. 22 is a cross-sectional plan view of the example power and liquid transfer assembly of FIG. 19.

(24) FIG. 23 is a view of the example power and liquid transfer assembly of FIG. 19.

(25) The figures are not to scale. Where appropriate, the same or similar reference numerals are used in the figures to refer to similar or identical elements.

DETAILED DESCRIPTION

(26) For the purpose of promoting an understanding of the principles of this disclosure, reference will be now made to the examples illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the claims is intended by this disclosure. Modifications in the illustrated examples and such further applications of the principles of this disclosure as illustrated therein are contemplated as would typically occur to one skilled in the art to which this disclosure relates.

(27) As used herein, the word “exemplary” means “serving as an example, instance, or illustration.” The embodiments described herein are not limiting, but rather are exemplary only. It should be understood that the described embodiments are not necessarily to be construed as preferred or advantageous over other embodiments. Moreover, the terms “embodiments of the invention,” “embodiments,” or “invention” do not require that all embodiments of the invention include the discussed feature, advantage, or mode of operation.

(28) As utilized herein the terms “circuits” and “circuitry” refer to physical electronic components (i.e. hardware) and any software and/or firmware (code) that may configure the hardware, be executed by the hardware, and or otherwise be associated with the hardware. As used herein, for example, a particular processor and memory may comprise a first “circuit” when executing a first set of one or more lines of code and may comprise a second “circuit” when executing a second set of one or more lines of code. As utilized herein, “and/or” means any one or more of the items in the list joined by “and/or”. As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. In other words, “x and/or y” means “one or both of x and y.” As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, “x, y, and/or z” means “one or more of x, y and z”. As utilized herein, the term “exemplary” means serving as a non-limiting example, instance, or illustration. As utilized herein, the terms “e.g.” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. As utilized herein, circuitry is “operable” to perform a function whenever the circuitry comprises the necessary hardware and code (if any is necessary) to perform the function, regardless of whether performance of the function is disabled or not enabled (e.g., by an operator-configurable setting, factory trim, etc.).

(29) As used herein, a wire-fed welding-type system refers to a system capable of performing welding (e.g., gas metal arc welding (GMAW), gas tungsten arc welding (GTAW), etc.), brazing, cladding, hardfacing, and/or other processes, in which a filler metal is provided by a wire that is fed to a work location, such as an arc or weld puddle.

(30) As used herein, a welding-type power source refers to any device capable of, when power is

applied thereto, supplying welding, cladding, plasma cutting, induction heating, laser (including laser welding and laser cladding), carbon arc cutting or gouging and/or resistive preheating, including but not limited to transformer-rectifiers, inverters, converters, resonant power supplies, quasi-resonant power supplies, switch-mode power supplies, etc., as well as control circuitry and other ancillary circuitry associated therewith.

(31) As used herein, preheating refers to heating the electrode wire prior to a welding arc and/or deposition in the travel path of the electrode wire.

(32) As used herein, the terms “front” and/or “forward” refer to locations closer to a welding arc, while “rear,” “back,” “behind,” and/or “backward” refers to locations farther from a welding arc.

(33) Some disclosed examples describe electric currents being conducted “from” and/or “to” locations in circuits and/or power supplies. Similarly, some disclosed examples describe “providing” electric current via one or more paths, which may include one or more conductive or partially conductive elements. The terms “from,” “to,” and “providing,” as used to describe conduction of electric current, do not necessitate the direction or polarity of the current. Instead, these electric currents may be conducted in either direction or have either polarity for a given circuit, even if an example current polarity or direction is provided or illustrated.

(34) Disclosed example welding assemblies for welding torches include: a first contact tip configured to conduct welding current to a consumable electrode; second contact tip configured to conduct preheating current to the consumable electrode; and a cooling body configured to transfer heat from at least the first contact tip to coolant and to conduct the welding current.

(35) Some example welding assemblies further include a contact tip insulator configured to insulate the second contact tip from the first contact tip, in which the contact tip insulator includes: an interior surface configured to be removably attached to a torch neck holding a contact tip; an exterior surface configured to contact the cooling body to prevent electrical contact between the cooling body and the second contact tip; and a seal on the exterior surface to substantially seal gas flow in cooperation with the cooling body. In some examples, the contact tip insulator comprises at least one of a ceramic or a plastic. In some examples, the contact tip insulator is a thermal insulator.

(36) Some example welding assemblies further include a wire guide, the wire guide comprising a dielectric material and having an inner bore configured to permit passage of a consumable electrode wire from the second contact tip to the first contact tip. In some examples, the dielectric material comprises at least one of a ceramic, a liquid ceramic, silicon nitride, or a dielectric exterior coating. In some examples, the second contact tip includes: an inner bore configured to deliver current to a consumable welding electrode; and screw threads on an exterior of the second contact tip, in which the screw threads include longitudinal slots to permit welding gas to flow along an exterior of the contact tip.

(37) In some examples, the second contact tip includes a hex head opposite the screw threads on an exterior of the preheating contact tip to enable attachment and detachment of the second contact tip via a tip of a welding assembly. Some example welding assemblies further include a diffuser including: a first exterior portion configured to be coupled to the cooling body; a first interior portion configured to receive welding gas from an interior of the cooling body; a second interior portion configured to receive the first contact tip and to conduct current to the first contact tip; and a gas diffusion port configured to conduct welding gas from the first interior portion to a second exterior portion, the gas diffusion port configured to conduct the welding gas in a radial direction of the diffuser. Some examples further include an insulator configured to electrically insulate the diffuser from a nozzle.

(38) In some example welding assemblies, the cooling body includes a liquid coolant inlet port, a liquid coolant outlet port, and a channel to channel the liquid coolant from the liquid coolant inlet port to the liquid coolant outlet port. In some examples, the liquid coolant inlet port includes a valve actuator configured to actuate a valve by insertion into a first socket, and the liquid coolant outlet port includes a valve actuator configured to actuate a valve by insertion into a second socket.

Some example welding assemblies further include an exterior nut configured to secure a first portion of the welding assembly to a second portion of the welding assembly, in which the first portion of the welding assembly includes the first contact tip. In some examples, the first portion of the welding assembly comprises at least one of the second contact tip or the cooling body. In some examples, the cooling body includes a conductive body having one or more channels between the liquid coolant inlet port and the liquid coolant outlet port, and a cover to enclose the one or more channels.

(39) Referring to FIG. 1, an example welding system **100** is shown in which a robot **102** is used to weld a workpiece **106** using a welding tool **108**, such as the illustrated bent-neck (i.e., gooseneck design) welding torch (or, when under manual control, a handheld torch), to which power is delivered by welding equipment **110** via conduit **118** and returned by way of a ground conduit **120**. The welding equipment **110** may comprise, inter alia, one or more power sources (each generally referred to herein as a “power supply”), a source of a shield gas, a wire feeder, and other devices. Other devices may include, for example, water coolers, fume extraction devices, one or more controllers, sensors, user interfaces, communication devices (wired and/or wireless), etc.

(40) The welding system **100** of FIG. 1 may form a weld (e.g., at weld joint **112**) between two components in a weldment by any known electric welding techniques. Known electric welding techniques include, inter alia, shielded metal arc welding (SMAW), MIG, flux-cored arc welding (FCAW), TIG, laser welding, sub-arc welding (SAW), stud welding, friction stir welding, and resistance welding. MIG, TIG, hot wire cladding, hot wire TIG, hot wire brazing, multiple arc applications, and SAW welding techniques, inter alia, may involve automated or semi-automated external metal filler (e.g., via a wire feeder). In multiple arc applications (e.g., open arc or sub-arc), the preheater may pre-heat the wire into a pool with an arc between the wire and the pool. The welding equipment **110** may be arc welding equipment having one or more power supplies, and associated circuitry, that provides a direct current (DC), alternating current (AC), or a combination thereof to an electrode wire **114** of a welding tool (e.g., welding tool **108**). The welding tool **108** may be, for example, a TIG torch, a MIG torch, or a flux cored torch (commonly called a MIG “gun”). The electrode wire **114** may be tubular-type electrode, a solid type wire, a flux-core wire, a seamless metal core wire, and/or any other type of electrode wire.

(41) As will be discussed below, the welding tool **108** employs a contact tip assembly that heats the electrode wire **114** prior to forming a welding arc using the electrode wire **114**. Suitable electrode wire **114** types includes, for example, tubular wire, metal cored wire, aluminum wire, solid gas metal arc welding (GMAW) wire, composite GMAW wire, gas-shielded FCAW wire, SAW wire, self-shielded wire, etc.

(42) In the welding system **100**, the robot **102**, which is operatively coupled to welding equipment **110** via conduit **118** and ground conduit **120**, controls the location of the welding tool **108** and operation of the electrode wire **114** (e.g., via a wire feeder) by manipulating the welding tool **108** and triggering the starting and stopping of the current flow (whether a preheat current and/or welding current) to the electrode wire **114** by sending, for example, a trigger signal to the welding equipment **110**. When welding current is flowing, a welding arc is developed between the electrode wire **114** and the workpiece **106**, which ultimately produces a weldment. The conduit **118** and the electrode wire **114** thus deliver welding current and voltage sufficient to create the electric welding arc between the electrode wire **114** and the workpiece **106**. At the point of welding between the electrode wire **114** and the workpiece **106**, the welding arc locally melts the workpiece **106** and electrode wire **114** supplied to the weld joint **112**, thereby forming a weld joint **112** when the metal cools.

(43) In certain aspects, in lieu of a robot **102**'s robotic arm, a human operator may control the location and operation of the electrode wire **114**. For example, an operator wearing welding headwear and welding a workpiece **106** using a handheld torch to which power is delivered by welding equipment **110** via conduit **118**. In operation, as with the system **100** of FIG. 1, an

electrode wire **114** delivers the current to the point of welding on the workpiece **106** (e.g., a weldment). The operator, however, could control the location and operation of the electrode wire **114** by manipulating the handheld torch and triggering the starting and stopping of the current flow via, for example, a trigger. A handheld torch generally comprises a handle, a trigger, a conductor tube, a nozzle at the distal end of the conductor tube, and a contact tip assembly. Applying pressure to the trigger (i.e., actuating the trigger) initiates the welding process by sending a trigger signal to the welding equipment **110**, whereby welding current is provided, and the wire feeder is activated as needed (e.g., to drive the electrode wire **114** forward to feed the electrode wire **114** and in reverse to retract the electrode wire **114**). Commonly owned U.S. Pat. No. 6,858,818 to Craig S. Knoener, for example, describes an example system and method of controlling a wire feeder of a welding-type system.

(44) FIG. 2 illustrates an example liquid-cooled welding torch **200**. The liquid-cooled welding torch **200** may be used to implement the welding tool **108** of FIG. 1 to deliver the electrode wire **114** to a workpiece while providing both resistive preheating power and welding-type power.

(45) The liquid-cooled welding torch **200** includes a welding assembly **202**, a mounting assembly **204**, a weld cable **206**, liquid cooling assemblies **208**, **210**, **212**, **214**, and a power and liquid transfer assembly **216**. As disclosed herein, the example liquid-cooled welding torch **200** may be used to replace conventional robotic welding torches with resistive preheating-enabled welding torches having a same tool center point (TCP). By replacing a torch with another torch having a same TCP, the robot may be capable of continuing a welding program using the replacement torch with little or no reprogramming of tool points.

(46) The welding assembly **202** receives weld current and preheating current, conducts the weld current to the electrode wire **114**, and conducts the preheating current through a portion of the electrode wire **114**. The example welding assembly **202** is liquid-cooled by liquid provided via the liquid cooling assemblies **208-214**. The example welding assembly **202** of FIG. 2 receives the weld current via one or more of the weld cable **206**, the liquid cooling assemblies **208** and **212**, and/or the liquid cooling assemblies **210** and **214**. Because the workpiece provides the return path for the weld current to the power supply, no return path is provided via the weld cable **206** or the liquid cooling assemblies **208**. The weld cable **206** is an air-cooled (or gas-cooled) cable. However, the weld cable **206** may also be liquid-cooled.

(47) The example welding assembly **202** receives the preheating current via the weld cable **206**, the liquid cooling assemblies **208** and **212**, and/or the liquid cooling assemblies **210** and **214**. In the example of FIG. 2, the weld current is conducted via a different one of the weld cable **206**, the liquid cooling assemblies **208** and **212**, or the liquid cooling assemblies **210** and **214** than the preheating current that has the same polarity (i.e., current flow direction). The welding assembly **202** conducts the preheating current through a section of the electrode wire **114** to heat the electrode wire via resistive heating (e.g., $I_{\text{sup}}^2 R$ heating). The preheat current then returns to a preheating power supply via a different one of weld cable **206**, the liquid cooling assemblies **208** and **212**, or the liquid cooling assemblies **210** and **214** to complete a preheating circuit.

(48) In the example of FIG. 2, the weld current path, the preheating current supply path, and the preheating current return path may all be different ones of the weld cable **206**, the liquid cooling assemblies **208** and **212**, and the liquid cooling assemblies **210** and **214**. In some examples, the weld current path may be superimposed with the preheating current supply path or the preheating current return path to reduce the net current in the conductor. For example, if the weld current is 300 A and the preheating current is 100 A, the weld current and the preheating current may be superimposed to result in a net current of 200 A.

(49) As described in more detail below, the welding assembly **202** and the liquid cooling assemblies **212**, **214** may be separated from the remainder of the liquid-cooled welding torch **200** via the power and liquid transfer assembly **216**.

(50) FIG. 3 illustrates an exploded view of the example liquid-cooled welding torch **200** of FIG. 2.

(51) As shown in FIG. 3, the example welding assembly **202** includes a nozzle **302**, a diffuser insulator **304**, a first contact tip **306**, a wire guide **308**, a gas diffuser **310**, a first contact tip insulator **312**, a second contact tip **314**, a second contact tip insulator **316**, a nozzle mount **318**, and a nozzle mount clamp **320**.

(52) A liquid-cooled power cable assembly **322** includes the liquid cooling assemblies **212**, **214**, a cooler **324**, and power connector pins **326**, **328**. The liquid-cooled power cable assembly **322** is described below in more detail with reference to FIGS. 6, 7A, 7B, and 8.

(53) The power and liquid transfer assembly **216** enables quick separation of the welding assembly **202** and the liquid-cooled power cable assembly **322** from the mounting assembly **204**. In the example of FIG. 3, the power and liquid transfer assembly **216** includes a saddle **330**, a saddle cover **332**, a saddle clasp **334**, and saddle clamps **336**. The power and liquid transfer assembly **216** is described below in more detail with reference to FIGS. 4 and 5.

(54) The example mounting assembly **204** includes a torch body **338**, a neck **340**, and a robot link assembly **342** including, for example, a bracket, a link arm, and a robot mounting disk.

(55) FIG. 4 is a more detailed depiction of an example liquid cooling assembly **400** that may be used to implement the liquid cooling assemblies **208**, **210** of FIG. 2. FIG. 5 is a cross-section view of the liquid cooling assembly **400** of FIG. 4.

(56) The liquid cooling assembly **400** includes a power cable socket **402** that holds a power transfer socket **404** and a liquid shutoff valve **406**. The liquid cooling assembly **400** also includes a hose **408** and an internal conductor **410**. The hose **408** is coupled to the power cable socket **402** on a first end and coupled to a power cable fitting **412** on a second end.

(57) The power cable socket **402** receives one of the power connector pins **326**, **328** to transfer cooling liquid and welding current and/or preheating current to a corresponding one of the liquid cooling assemblies **212**, **214**. The power transfer socket **404** enables insertion of the power connector pin **326**, **328**, and transfers current to and/or from an inserted power connector pin **326**, **328**. An example power transfer socket that may be used to implement the power transfer socket **404** is a PowerBud® power contact, sold by Methode Electronics, Inc., which provides multiple contact points between the power transfer socket and an inserted power connector pin **326**, **328**.

(58) The liquid shutoff valve **406** selectively permits flow of liquid from the hose **408** to the power transfer socket **404** and to a connected liquid cooling assembly **212**, **214**. The example liquid shutoff valve **406** is a Schrader valve. However, other types of valves may be used to implement the liquid shutoff valve **406**. When a power connector pin **326**, **328** is inserted (e.g., fully inserted) into the power transfer socket **404**, the power connector pin **326**, **328** displaces (e.g., unseats) a stem **414** from a core **416** of the valve **406**, which permits liquid to flow to and/or from the hose **408**. When the power connector pin **326**, **328** is removed or partially removed, the stem **414** is forced back into the core **416** and stops flow of liquid.

(59) The hose **408** is coupled to the power cable fitting **412** via a ferrule **418** and a fitting nut **420**. The power cable fitting **412** is coupled to a source of weld current and/or preheating current, and is electrically connected to transfer the weld current and/or preheating current to or from the internal conductor **410** of the hose **408**. The hose **408** is also coupled to the power cable socket **402** via a ferrule **418**. The example power cable fitting **412**, the example power cable socket **402**, and/or the hose **408** include hose barbs to secure the hose **408**. However, other methods of securing the hose to the power cable fitting **412** and/or the power cable socket **402** may be used, such as clamps, compression fittings, or any other hose fittings.

(60) During operation, when the power cable fitting **412** is coupled to a power source and a liquid cooling device, and the power cable socket **402** is coupled to a liquid cooling assembly **212**, **214**, the example liquid cooling assembly **400** permits liquid to flow through the power cable fitting **412**, the hose **408**, the valve **406**, the power cable socket **402**, and the power transfer socket **404**, either to or from the liquid cooling device (e.g., based on whether the assembly is configured as the liquid supply or the liquid return). The example liquid cooling assembly **400** also conducts current

from and/or to a weld power supply and/or a preheating power supply. For example, the current is conducted through the power cable fitting **412**, the internal conductor **410**, the power cable socket **402**, and the power transfer socket **404**.

(61) FIG. **6** is an exploded view of the example liquid-cooled power cable assembly **322** of FIG. **3**. The liquid-cooled power cable assembly **322** provides cooling liquid supply and return paths from the liquid cooling assemblies **212**, **214** to the welding assembly **202** of FIG. **3**. The liquid-cooled power cable assembly **322** includes the liquid cooling assemblies **212**, **214**, the power connector pins **326**, **328**, a cooling body **602**, and a cooling body cover **604**.

(62) Each of the example liquid cooling assemblies **212**, **214** includes three layers: an inner conductive layer **606a**, **606b**; an insulative layer **608a**, **608b**; and an outer protective layer **610a**, **610b**. The inner conductive layer **606a**, **606b** conduct current and liquid, and is constructed of a conductive material such as copper. The insulative layers **608a**, **608b** provide electrical insulation between the inner conductive layers **606a**, **606b** and the outer protective layers **610a**, **610b**. The example insulative layers **608a**, **608b** may be silicone, PTFE, PET, and/or any other suitable electrically insulative material or combination of materials. The outer protective layers **610a**, **610b** provide rigidity and/or physical protection from damage, such as punctures. The outer protective layers **610a**, **610b** may be a rigid material such as aluminum or any other appropriate material or combination of materials.

(63) In some examples, two or more of the layers **606a-610a**, **606b-610b** may be combined. For example, the insulative layers **608a**, **608b** may also serve as the outer protective layers **610a**, **610b**, or vice versa. In other examples, the outer protective layers **610a**, **610b** may be omitted.

(64) The inner conductive layers **606a**, **606b** are contained within the insulative layers **608a**, **608b**. The insulative layers **608a**, **608b** are similar contained within the outer protective layers **610a**, **610b**.

(65) One of the liquid cooling assemblies **212**, **214** supplies cooling liquid to the cooling body **602**, and the other of the liquid cooling assemblies **212**, **214** receives the liquid from the cooling body **602**. The cooling body **602** circulates the liquid through a tortuous path **612** between a liquid input port and a liquid output port. The cooling body **602** is coupled to the welding assembly **202** to conduct heat from the components in the welding assembly **202** to the liquid, thereby cooling the welding assembly **202**. The cooling body cover **604** is attached to the cooling body **602** to contain the fluid within the tortuous path **606**. In some examples, the cooling body **602** and a cooling body cover **604** may be a single unit (e.g., constructed using additive manufacturing techniques).

(66) FIGS. **7A**, **7B**, and **7C** are views of the example cooler body **602** of FIG. **6**. As illustrated in FIGS. **7A**, **7B**, and **7C**, the tortuous path **612** includes a continuous path around a circumference of the cooler body **602**. In the illustrated example, the liquid cooling assembly **214** supplies cooling liquid to the cooler body **602** and the liquid cooling assembly **212** returns the cooling liquid to a liquid cooler. The cooling liquid follows a flow path **702** (shown using a dotted line) through the tortuous path **612** to increase heat transfer from the welding assembly **202** to the liquid.

Continuities A and B from one view to the next are shown in FIGS. **7A**, **7B**, and **7C**.

(67) FIG. **8** is a cross-section view of the liquid cooling assembly **212**, **214** and the power and liquid transfer assembly **216** of FIG. **2** coupled to the liquid cooling assembly **400** of FIG. **4**.

(68) As illustrated in FIG. **8**, the power cable socket **402** is seated in the saddle **330**, which prevents movement of the power cable socket **402**. The example power connector pin **326** is inserted into the power cable socket **402** and the power transfer socket **404**. The power connector pin **326** includes an inner liquid coolant passage **802**, one or more liquid coolant ports **804** to enable liquid to flow into and/or out of the liquid coolant passage **802**, and a seal **806**. When inserted, the power connector pin **326** opens the valve **406** to permit liquid to flow through the valve **406**, the liquid coolant ports **804**, the liquid coolant passage **802**, and the inner conductive layer **606a**. A retaining ring **808** may be included in the saddle cover **332** to hold the power connector pin **326** in place.

(69) In addition to placing the liquid cooling assembly **212** in fluid communication with the liquid

cooling assembly **208**, the example power connector pin **326** also conducts weld current and/or preheating current between the liquid cooling assembly **212** and the liquid cooling assembly **208**. The example inner conductive layer **606a** is in electrical contact with the power transfer pin **326**, which is a conductive material (e.g., copper) and is in electrical contact with the power transfer socket **404**.

(70) While the examples are described with reference to the liquid cooling assembly **208**, the liquid cooling assembly **212**, and the power connector pin **326**, these examples are similarly applicable to the liquid cooling assembly **210**, the liquid cooling assembly **214**, and the power connector pin **328**.

(71) FIG. **9** illustrates the example welding assembly **202** and the example liquid cooling assemblies **212**, **214** of FIG. **2** disconnected from the remainder of the torch **200** via the power and liquid transfer assembly **216**. For detachment of the example liquid cooling assemblies **212**, **214**, the example saddle clasp **334** is unhooked or otherwise detached from a retention pin **902** on the saddle **330**. When the saddle clasp **334** is unhooked, the power connector pin **328** can be disengaged from the power cable socket **402** and the saddle cover **332** can be simultaneously lifted from the saddle **330**. Conversely, to install the liquid cooling assemblies **212**, **214**, the power connector pins **326**, **328** are inserted into corresponding power cable sockets **402** while the saddle cover **332** is placed onto the saddle **330**. When the power connector pins **326**, **328** and the saddle cover **332** are in place on the saddle **330**, the saddle clasp **334** is hooked into the retention pin **902** to hold the power connector pins **326**, **328** in place.

(72) FIG. **10** is a cross-section view of the example welding assembly **202** of FIG. **2**. The welding assembly **202** is liquid-cooled via the liquid cooling assemblies **212**, **214**. The liquid cooling assemblies **212**, **214** and/or a torch neck **1002** provide weld current and preheating current to the welding assembly **202** for preheating the electrode wire **114** and for generating a welding arc.

(73) The example welding assembly **202** includes the nozzle **302**, the diffuser insulator **304**, the first contact tip **306**, the wire guide **308**, the gas diffuser **310**, the first contact tip insulator **312**, the second contact tip **314**, the second contact tip insulator **316**, the nozzle mount **318**, the nozzle mount clamp **320**, the cooling body **602**, and the cooling body cover **604**. The welding assembly **202** is attached to a torch neck **1002**, through which a wire liner **1004** conveys the electrode wire **114** and/or shielding gas to the welding assembly **202**.

(74) The first contact tip **306** delivers welding current to the electrode wire **114** for arc welding. The first contact tip **306** is threaded into a gas diffuser **310**, which is in turn threaded into the diffuser insulator **304**. The diffuser insulator **304** provides electrical and thermal insulation between the gas diffuser **310** and the nozzle **302**.

(75) The gas diffuser **310** is threaded into the cooling body **602**. The cooling body **602** conducts welding current and/or preheating current from the liquid-cooled power cable assembly **322** (e.g., from the inner conductive layer(s) **606a**, **606b**) to the diffuser **310**, which is electrically connected to the first contact tip **306**. The first contact tip insulator **312** and the diffuser insulator **304** provide electrical insulation between the weld current and preheat current path(s) and the nozzle **302**.

(76) The second contact tip **314** is electrically coupled to the torch neck **1002** to conduct preheating current to and/or from the electrode wire **114**. The preheating circuit includes the torch neck **1002**, the second contact tip **314**, the first contact tip **306**, a portion of the electrode wire **1006** between the second contact tip **314** and the first contact tip **306**, the diffuser **310**, the cooling body **602**, and one or both of the inner conductive layers **606a**, **606b** in the liquid-cooled power cable assembly **322**.

(77) The second contact tip insulator **316** provides electrical insulation between the second contact tip **314** and the cooling body **602**. The second contact tip insulator **316** includes a seal **1008** (e.g., an o-ring) to reduce or prevent welding gas leakage.

(78) The nozzle mount **318** and the nozzle mount clamp **320** provide an attachment point for threading the welding assembly **202** onto the torch neck **1002**. The nozzle mount **318** physically

couples and/or provides support to the liquid-cooled power cable assembly **322** from the torch neck **1002**, which is rigid.

(79) In addition to the welding assembly **202**, the liquid-cooled power cable assembly **322**, and the torch neck **1002** being detachable from the mounting assembly **204** (e.g., via the power and liquid transfer assembly **216** and a conventional disconnection feature between the torch neck **1002** and the mounting assembly **204**), the welding assembly **202** may be completely or partially disassembled to access one or more of the components in the welding assembly **202**.

(80) In the example of FIG. **10**, the first contact tip **306**, the wire guide **308**, and/or the second contact tip **314** are removable via the tip of the nozzle **302**. FIGS. **11A**, **11B**, and **11C** show an example implementation of the first contact tip **306**, FIGS. **12A** and **12B** show an example implementation of the wire guide **308**, and FIGS. **13A**, **13B**, and **13C** show an example implementation of the second contact tip **314**.

(81) As shown in FIGS. **10** and **11A**, a first end **1102** of the first contact tip **306** has a hexagonal cross-section. The hexagonal cross-section enables the first contact tip **306** to be unthreaded from the diffuser **310** via the opening in the nozzle **302**. Other exterior geometries may be used for the cross-section of the first end **1102** of the first contact tip **306**. Additionally or alternatively, an interior geometry may be used (e.g., in combination with a corresponding tool) to unthread the first contact tip **306** from the diffuser **310**.

(82) After removal of the first contact tip **306** from the welding assembly **202** via the nozzle **302**, the wire guide **308** may also be removed via the nozzle **302**. As shown in FIGS. **10**, **12A**, and **12B**, an outer surface **1202** of the wire guide **308** is relatively smooth (e.g., not threaded) and can be inserted into and removed from of an inner diameter **1010** of the cooling body **602** without threading. The wire guide **308** has a wire path **1204** to guide the wire from the second contact tip **314** to the first contact tip **306**. In some examples, the wire guide **308** is a nonconductive material such as ceramic, so that the electrode wire **114** is the only conductive path between the second contact tip **314** and the first contact tip **306**.

(83) As shown in FIGS. **13A**, **13B**, and **13C**, the second contact tip **314** includes a hexagonal cross-section on a first end **1302** of the second contact tip **314**. An interior surface **1304** and/or an exterior surface **1306** (e.g., a head) of the first end **1302** may provide the hexagonal cross-section and/or any other shape that enables the second contact tip to be unthreaded from the torch neck **1002**. The example second contact tip **314** includes threads **1305** to enable the second contact tip **314** to be threaded into the second contact tip insulator **316**. The exterior surface **1306** has a smaller diameter than the major diameter of the threads **1305**, which can improve access to the exterior surface **1306** for removal and/or installation of the second contact tip **314**.

(84) As shown in FIG. **10**, the outer diameter of the second contact tip **314** is equal to or less than the inner diameter **1010** of the cooling body **602** and less than an inner diameter **1012** of the second contact tip insulator **316** to enable the second contact tip **314** to be removed via the tip of the nozzle **302**, the cooling body **602**, and the second contact tip insulator **316**. The first end **1302** may be manipulated (e.g., via a tool inserted through the nozzle **302**) to unthread the second contact tip **314** from the torch neck **1002**, after which the second contact tip **314** can be removed from the welding assembly **202** without disassembly of the welding assembly **202**.

(85) As shown in FIGS. **13A**, **13B**, and **13C**, the second contact tip **314** includes slots **1308** running longitudinally on the exterior of the second contact tip **314** through the threads **1305**. The slots **1308** permit the flow of welding gas from the interior of the torch neck **1002** to the diffuser **310** via the cooling body **602** (e.g., through the interior of the cooling body **602** through and/or around the wire guide **308**).

(86) The second contact tip **314** includes a first bore **1310** having a first diameter. The electrode wire **114** makes electrical contact with the first bore **1310**. The example second contact tip **314** may also have a second, larger bore **1312** on a rear side of the contact tip **314** from the first bore **1310**. The length of the second bore **1312** may be selected to control a contact length of the first bore

1310.

(87) As illustrated in FIG. 10, the example contact tip **314** may have an insert **1012** inserted within the first bore **1310** to improve contact with the electrode wire **114**, improve current transfer with the electrode wire **114**, and/or improve mechanical strength at operating temperatures.

(88) FIGS. 14A and 14B illustrate views of the example diffuser **310** of FIGS. 3 and 10. As discussed above with reference to FIG. 10, the diffuser **310** includes two sets of exterior threads for installation into the diffuser insulator **304** and the cooling body **602**, and a set of interior threads for installation of the first contact tip **306** into the diffuser **310**. The diffuser **310** includes gas outlets **1402** to enable the flow of gas from the inner diameter **1010** of the cooling body **602** to the nozzle **302** for output to the weld.

(89) FIG. 15A illustrates a conventional robotic welding torch **1500** having a first tool center point distance and torch neck angle. FIG. 15B illustrates an example implementation of the liquid-cooled welding torch **200** of FIG. 2 configured to replace the conventional torch **1500** of FIG. 15A while maintaining a same tool center point distance and torch neck angle.

(90) The tool center point distance **1502** of the conventional torch **1500** is a function of a stickout distance **1504**, a nozzle length **1506**, a nozzle angle **1508**, a neck bend angle **1510**, a tool center point area **1512**, and a spacer width **1514**. The tool center point area **1512** (e.g., Area.sub.TCP) is defined using Equation 1 below:

$$\text{Area.sub.TCP} = 0.5 * ((\text{TCP} - \text{spacer width } 1514) \cdot \tan(\text{nozzle angle } 1508) + ((\text{TCP} - \text{spacer width } 1514) - (\text{nozzle length } 1506 + \text{stickout } 1504) \cdot \cos(\text{nozzle angle } 1508)) \cdot \tan(\text{nozzle angle } 1508 - \text{neck bend angle } 1510) - 1))$$
 (Equation 1)

(91) The example welding assembly **202** described above with reference to FIGS. 10, 11A-11C, 12A, 12B, 13A-13C, 14A, and/or 14B may have a same nozzle length **1506** as conventional nozzles. In other examples, the welding assembly **202** may require a longer nozzle **302** than a nozzle of the conventional torch. In some examples, such as the welding torch described below with reference to FIGS. 19-23, the welding assembly may have a nozzle that is shorter than conventional nozzles. However, the length of the example torch neck **1002** and/or the length and/or one or more dimensions of the example mounting assembly **204** may be adjusted to compensate for differences in the nozzle length.

(92) Using the example welding assembly **202** disclosed herein, the example liquid-cooled welding torch **200** may be dimensioned to be a replacement for any standard tool center point distance (e.g., 350 mm, 400 mm, 450 mm, 500 mm, etc.) and/or torch neck angle (e.g., 0 degree, 22 degree, 35 degree, 45 degree, etc.) to maintain the same TCP, torch neck angle, and tool center point area after replacement. In other words, disclosed example welding assemblies **202** and/or liquid-cooled welding torches **200** may be used as replacements for conventional robotic weld torches such that a robot on which the replacement occurs does not require reprogramming or recalibration of new tool center point(s) or torch neck angle(s). After replacement of the conventional welding torch with disclosed example liquid-cooled welding torches, the robot subject to replacement is capable of higher deposition rates, improved welding starts, and/or other advantages over conventional welding torches. As a result, reprogramming of welding voltages, currents, and/or torch travel speeds may be performed to realize the advantages of the liquid-cooled welding torch **200** for previously programmed welding tasks.

(93) In addition to the tool center point distance **1502**, the stickout distance **1504**, the nozzle length **1506**, the nozzle angle **1508**, the neck bend angle **1510**, the tool center point area **1512**, and the spacer width **1514** dimensions of the conventional welding torch **1500**, the example welding torch **200** includes a preheat distance **1516** within the nozzle. The nozzle length **1506** is subdivided into the stickout distance **1504**, the preheat distance **1516**, and a neck-to-contact tip length **1518**. The conventional torch **1500** may be considered to have a preheat distance **1516** of 0. By replacing a conventional weld torch **1500** with a weld torch **200** having substantially the same tool center point distance **1502**, the stickout distance **1504**, the nozzle length **1506**, the nozzle angle **1508**, the neck

bend angle **1510**, and the tool center point area **1512**, the replacement reduces the programming needed to avoid an increased risk of collisions.

(94) Tables 1 and 2 below illustrates example comparisons of the dimensions **1502-1516** for two example tool center point distances **1502**, 350 mm and 400 mm.

(95) TABLE-US-00001 TABLE 1 350 mm TCP Dimensions for Conventional and Disclosed example welding torches Neck-to- Nozzle Nozzle Neck TCP TCP Stickout Preheat contact tip length angle angle TCP area # (mm) (in) (in) (in) length (in) (in) (deg.) (deg.) (sq. in.) 1 350 13.78 0.59 0.00 4.33 4.92 45 22 51.59 2 350 13.78 0.75 3.29 1.78 5.82 45 22 54.61 3 350 13.78 0.75 2.40 1.78 4.93 45 22 51.61

(96) TABLE-US-00002 TABLE 2 400 mm TCP Dimensions for Conventional and Disclosed example welding torches Neck-to- Nozzle Nozzle Neck TCP TCP Stickout Preheat contact tip length angle angle TCP area # (mm) (in) (in) (in) length (in) (in) (deg.) (deg.) (sq. in.) 4 400 15.75 0.59 0.00 4.33 4.92 45 22 66.44 5 400 15.75 0.75 3.29 1.78 5.82 45 22 70.17 6 400 15.75 0.75 2.40 1.78 4.93 45 22 66.46

(97) A welding torch having a preheat length of 3.29 inches (e.g., torches **2** and **5** of Tables 1 and 2 above) may be used with an air-cooled torch. A welding torch having a preheat length of 2.40 inches results in a nozzle length **1506** and TCP area **1512** of a welding torch **200** that substantially match the nozzle length and TCP area **1512** of the conventional torch **1500**. However, the current density may increase enough to require an increase in the radius of the welding assembly **202** (e.g., increased components in the welding assembly **202**) to continue to use air cooling (as in torches **2** and **5** of Tables 1 and 2). Additionally or alternatively, the preheat distance of 2.40 inches may be used with the example liquid cooled welding torch **200** disclosed herein.

(98) Any of the dimensions of Tables 1 and 2 (or similar dimensions) may be selected based on a specification of another of the dimensions. For example, the preheating length (and, thus, the dimensions of the welding assembly) may be selected based on any of the TCP, the stickout, the neck-to-contact tip length, the nozzle length, the nozzle angle, the neck angle, and/or the TCP area. Additionally or alternatively, the neck-to-contact tip length is based on at least one of a nozzle length, a preheat distance, a nozzle angle, or a neck angle.

(99) Preheat lengths **1516** that are longer than the 3.29 inches of the examples of Tables 1 and 2 may result in a torch envelope that is larger than the envelope of the conventional torches **1500** by more than an acceptable amount. Torches having an envelope that is excessively large may require more robot programming to serve as a replacement for a conventional torch and/or has a higher chance of collision during operation. Additional programming and collisions are undesirable effects of replacement, and disclosed examples reduce or prevent such effects while providing the benefits of electrode preheating at the welding assembly **202**.

(100) While the example welding assembly **202** includes the first contact tip **306** and the second contact tip **314** such that both of the contact tips **306**, **314** are on a distal end of the bend in the torch neck **1002**, in some other examples the second contact tip **314** is on a proximal side of the bend in the torch neck **1002**. Additionally or alternatively, a third contact tip is further located on the proximal side of the bend in the torch neck **1002** so that preheating occurs between the second contact tip **314** (on either side of the bend) and the third contact tip (on the proximal side of the bend).

(101) When the electrode wire **114** is heated beyond a particular temperature, the column strength of the electrode wire **114** may be reduced to a point that the heated electrode wire **114** can no longer be pushed around the bend in the torch neck **1002** without buckling. In some such examples, the liquid-cooled welding torch **200** is provided with a pull motor to assist the push motor located in the wire feeder.

(102) The bend in the torch neck **1002** may be provided with ceramic bearings to reduce the friction force between the wire liner **1004** and the electrode wire **114**, which increases the temperature to which the electrode wire **114** may be preheated before buckling becomes likely.

(103) In some examples, the length between the first and second contact tips **306, 314** (e.g., the preheat length) is adjustable to change the length of the electrode wire **114** that is being preheated. As the preheat length increases, the energy added to the electrode wire **114** is increased per unit of current. To change the preheat length, one or both of the contact tips **306, 314** may be configured to be translated in the axial direction via manual and/or automatic methods.

(104) Using the second contact tip **314** as an example, the second contact tip **314** may be threaded into an intermediate device between the second contact tip **314** and the torch neck **1002**. The intermediate device may be automatically rotated (e.g., with a motor coupled to the contact tip) and/or manually rotated (e.g., with a thumb wheel or other device configured to cause rotation in the contact tip) while limiting rotation of the second contact tip **314**, which causes the threads of the second contact tip **314** to carry the second contact tip **314** toward or away from the first contact tip **306**.

(105) Additionally or alternatively, the first and/or second contact tips **306, 314** may be configured to be reversible to change the preheat length. For example, if the first contact tip **306** has a contact location with the electrode wire **114** that is closer than a midpoint of the first contact tip **306** to the second contact tip **314**, reversing the first contact tip **306** changes the contact location with the electrode wire **114** and extends the preheat length. In some examples, different contact tips have different contact points, so that changing contact tips changes the preheat length. In some other examples, the welding assembly **202** may be replaced with a different welding assembly that has a different preheat length such as different spacing between the contact tips **306, 314** (which may require a different robotic program to be run that accounts for a different nozzle length).

(106) The change in preheat length may be automatically controlled based on, for example, a temperature of the electrode wire **114** and/or based on a desired preheat level or heat input to the weld specified by a user. In some examples, a current controlled control loop is used to control preheat current when the preheat length is automatically adjustable.

(107) The power and liquid transfer assembly **216** is attached to the torch body **338** on a side of the torch body **338** opposite the mounting assembly **204**, or opposite a direction of the bend in the torch neck **1002**. Because the power and liquid transfer assembly **216** increasing a volume of the torch **200**, locating the power and liquid transfer assembly **216** opposite the mounting assembly **204** and/or opposite a direction of the bend in the torch neck **1002** may reduce the chances of collision with a workpiece when using the same program with the replacement preheating weld torch **200** as used with a conventional weld torch.

(108) In still other examples, the welding assembly **202** may be provided with a wire oscillator to cause physical oscillation or weave at a tip of the electrode wire **114**. An example implementation of the wire oscillator that may be used to provide wire oscillation is described in U.S. Pat. No. 4,295,031. Using both the wire oscillator and the contact tips **306, 314**, disclosed example welding torches may provide both wire oscillation and resistive preheating to a weld to further improve deposition rates and weld quality.

(109) FIG. **16** is a cross section of an example welding cable **1600** that may be used to provide cooling liquid, welding current, and preheating current to a welding torch, and to carry cooling liquid away from the welding torch. The example welding cable **1600** may be used instead of the liquid cooling assemblies **208, 210**, in cases in which the first contact tip **306** is part of the preheating circuit and the welding circuit.

(110) The welding cable **1600** is a coaxial-type welding cable, in which the electrode wire **114** is fed through a wire guide **1602**. The wire guide **1602** is surrounded by a jacket **1604**. A first conductor **1606** provides a first electrical path for preheating current, or welding current and preheating current. The first conductor **1606** may be, for example, a copper sheath or webbing rated to conduct welding current. The first conductor **1606** is surrounded by a jacket **1608**.

(111) A second conductor **1610** provides a second electrical path for preheat current, or welding current and preheating current. In the example of FIG. **16**, a first one of the first conductor **1606** or

the second conductor **1610** provides a path for the preheating current and is coupled to a preheating power supply. The second conductor **1610** is surrounded by a jacket **1612**. In some examples, another protective layer may be present outside of the jacket **1612** to protect the welding cable **1600** from damage. In some examples, the jacket **1612** provides both physical and electrical protection to the cable **1600**.

(112) An annulus **1614** between the jacket **1604** and the first conductor **1606** and/or the jacket **1608** conducts cooling liquid in a first direction (e.g., from a liquid cooler to the welding torch, from the welding torch to the liquid cooler). An annulus **1616** between the jacket **1608** and the second conductor **1610** and/or the jacket **1612** conducts the cooling liquid in a second direction opposite the liquid flow direction in the annulus **1614**.

(113) FIG. 17 illustrates a functional diagram of an example welding system **1700** including the example welding torch **200** of FIG. 2, and which may be used with the welding system **100** of FIG. 1. The welding system **1700** includes the weld torch **200** having the first contact tip **306** and a second contact tip **314**. The system **1700** further includes the electrode wire **114** fed from a wire feeder **1702** having a wire drive **1704** and a wire spool **1706**, a preheating power supply **1708**, and a welding power supply **1710**. The system **1700** is illustrated in operation as producing a welding arc **1712** between the electrode wire **114** and a workpiece **106**.

(114) In operation, the electrode wire **114** passes from the wire spool **1706** through the second contact tip **314** and the first contact tip **306**, between which the preheating power supply **1708** generates a preheating current to heat the electrode wire **114**. Specifically, in the configuration shown in FIG. 17, the preheating current enters the electrode wire **114** via the second contact tip **314** (e.g., via the wire drive **1704** and/or the torch neck **1002** of FIG. 10) and exits via the first contact tip **306**. At the first contact tip **306**, a welding current may also enter the electrode wire **114** (e.g., via the liquid cooling assemblies **208** and **212**, and/or **210** and **214**). The welding current is generated, or otherwise provided by, the welding power supply **1710**. The welding current exits the electrode wire **114** via the workpiece **106**, which in turn generates the welding arc **1712**. When the electrode wire **114** makes contact with a target metal workpiece **106**, an electrical circuit is completed and the welding current flows through the electrode wire **114**, across the metal work piece(s) **106**, and returns to the welding power supply **1710**. The welding current causes the electrode wire **114** and the parent metal of the work piece(s) **106** in contact with the electrode wire **114** to melt, thereby joining the work pieces as the melt solidifies. By preheating the electrode wire **114**, a welding arc **1712** may be generated with drastically reduced arc energy. Generally speaking, the preheating current is proportional to the distance between the contact tips **306**, **314** and the electrode wire **114** size.

(115) The welding current is generated, or otherwise provided by, a welding power supply **1710**, while the preheating current is generated, or otherwise provided by, the preheating power supply **1708**. The preheating power supply **1708** and the welding power supply **1710** may ultimately share a common power source (e.g., a common generator or line current connection), but the current from the common power source is converted, inverted, and/or regulated to yield the two separate currents—the preheating current and the welding current. For instance, the preheat operation may be facilitated with a single power source and associated converter circuitry, in which case three leads may extend from a single power source.

(116) During operation, the system **1700** establishes a welding circuit to conduct welding current from the welding power supply **1710** to the first contact tip **306** via the liquid cooling assemblies **208** and **212** and/or **210** and **214**, and returns to the power supply **1710** via the welding arc **1712**, the workpiece **106**, and a work lead **1718**. To enable connection between the welding power supply **1710** and the first contact tip **306** and the workpiece **106**, the welding power supply **1710** includes terminals **1720**, **1722** (e.g., a positive terminal and a negative terminal).

(117) During operation, the preheating power supply establishes a preheating circuit to conduct preheating current through a section **1726** of the electrode wire **114**. To enable connection between

the preheating power supply **1708** and the contact tips **306**, **314**, the preheating power supply **1708** includes terminals **1728**, **1730**. The preheating current flows from the welding power supply **1710** to the second contact tip **314** (e.g., via the torch neck **1002**), the section **1726** of the electrode wire **114**, the first contact tip **306**, and returns to the preheating power supply **1708** via a cable **1732** connecting the terminal **1720** of the welding power supply **1710** to the terminal **1730** of the preheating power supply **1708**.

(118) Because the preheating current path is superimposed with the welding current path over the connection between the first contact tip **306** and the power supplies **1708**, **1710**, the cable **1732** may enable a more cost-effective single connection between the first contact tip **306** and the power supplies **1708**, **1710** (e.g., a single cable) than providing separate connections for the welding current to the first contact tip **306** and for the preheating current to the first contact tip **306**. In other examples, the terminal **1730** of the preheating power supply **1708** is connected to the first contact tip **306** via a separate path than the path between the first contact tip **306** and the welding power supply **1710**. For example, the welding current may be conducted via the liquid cooling assemblies **208** and **212** while the preheating current is conducted via the liquid cooling assemblies **210** and **214** (or vice versa).

(119) In some examples, the welding torch may include a push-pull wire feed system by including a feed motor located at or near the weld torch to pull the electrode wire **114**. In some examples, the inclusion of the pulling feed motor enables the portion of the electrode wire **114** that is preheated to a different location along the wire feed path than in the examples of FIGS. **2**, **3**, and **10**. For example, the second contact tip **314** may be moved into the neck **1002** (e.g., prior to the bend in the torch neck **1002** in the feed direction of the electrode wire **114**) and/or in the torch body (e.g., the mounting assembly **204** of FIG. **2**) and/or multiple contact tips may be positioned at locations along the length of the electrode wire **114** to provide a preheating circuit that is separate from the welding circuit (e.g., does not share a same contact tip with the welding circuit) and/or provides an additional preheating circuit (e.g., a first preheating current applied to a first portion of the electrode wire **114** and a second preheat current applied to a second portion of the preheating wire). In some examples, the idler roller of a push-pull wire feed system may function as a contact tip to conduct preheating current. By moving all or a portion of the preheating circuit to the wire source side of the bend in the torch neck **1002** (e.g., the side of the bend closer to the wire spool in the electrode feed path), the size of the welding assembly may be reduced, the preheat length may be increased, and/or the preheating current may be reduced. Reduction in the size of the welding assembly and reduction in the preheating current enables torch dimensions that are closer to those of conventional, non-resistive preheating torches, further improving the ease of replacement of conventional torches with torches providing resistive preheating.

(120) FIG. **18** is a block diagram of an example implementation of the power supplies **1708**, **1710** of FIG. **17**. The example power supply **1708**, **1710** powers, controls, and supplies consumables to a welding application. In some examples, the power supply **1708**, **1710** directly supplies input power to the welding tool **108**. In the illustrated example, the power supply **1708**, **1710** is configured to supply power to welding operations and/or preheating operations. The example power supply **1708**, **1710** also provides power to a wire feeder to supply the electrode wire **114** to the welding tool **108** for various welding applications (e.g., GMAW welding, flux core arc welding (FCAW)).

(121) The power supply **1708**, **1710** receives primary power **1808** (e.g., from the AC power grid, an engine/generator set, a battery, or other energy generating or storage devices, or a combination thereof), conditions the primary power, and provides an output power to one or more welding devices and/or preheating devices in accordance with demands of the system. The primary power **1808** may be supplied from an offsite location (e.g., the primary power may originate from the power grid). The power supply **1708**, **1710** includes a power converter **1810**, which may include transformers, rectifiers, switches, and so forth, capable of converting the AC input power to AC and/or DC output power as dictated by the demands of the system (e.g., particular welding

processes and regimes). The power converter **1810** converts input power (e.g., the primary power **1808**) to welding-type power based on a weld voltage setpoint and outputs the welding-type power via a weld circuit.

(122) In some examples, the power converter **1810** is configured to convert the primary power **1808** to both welding-type power and auxiliary power outputs. However, in other examples, the power converter **1810** is adapted to convert primary power only to a weld power output, and a separate auxiliary converter is provided to convert primary power to auxiliary power. In some other examples, the power supply **1708, 1710** receives a converted auxiliary power output directly from a wall outlet. Any suitable power conversion system or mechanism may be employed by the power supply **1708, 1710** to generate and supply both weld and auxiliary power.

(123) The power supply **1708, 1710** includes a controller **1812** to control the operation of the power supply **1708, 1710**. The power supply **1708, 1710** also includes a user interface **1814**. The controller **1812** receives input from the user interface **1814**, through which a user may choose a process and/or input desired parameters (e.g., voltages, currents, particular pulsed or non-pulsed welding regimes, and so forth). The user interface **1814** may receive inputs using any input device, such as via a keypad, keyboard, buttons, touch screen, voice activation system, wireless device, etc. Furthermore, the controller **1812** controls operating parameters based on input by the user as well as based on other current operating parameters. Specifically, the user interface **1814** may include a display **1816** for presenting, showing, or indicating, information to an operator. The controller **1812** may also include interface circuitry for communicating data to other devices in the system, such as the wire feeder. For example, in some situations, the power supply **1708, 1710** wirelessly communicates with other welding devices within the welding system. Further, in some situations, the power supply **1708, 1710** communicates with other welding devices using a wired connection, such as by using a network interface controller (NIC) to communicate data via a network (e.g., ETHERNET, 10baseT, 10base100, etc.). In the example of FIG. **18**, the controller **1812** communicates with the wire feeder via the weld circuit via a communications transceiver **1818**.

(124) The controller **1812** includes at least one controller or processor **1820** that controls the operations of the welding power supply **1802**. The controller **1812** receives and processes multiple inputs associated with the performance and demands of the system. The processor **1820** may include one or more microprocessors, such as one or more “general-purpose” microprocessors, one or more special-purpose microprocessors and/or ASICS, and/or any other type of processing device. For example, the processor **1820** may include one or more digital signal processors (DSPs).

(125) The example controller **1812** includes one or more storage device(s) **1823** and one or more memory device(s) **1824**. The storage device(s) **1823** (e.g., nonvolatile storage) may include ROM, flash memory, a hard drive, and/or any other suitable optical, magnetic, and/or solid-state storage medium, and/or a combination thereof. The storage device **1823** stores data (e.g., data corresponding to a welding application), instructions (e.g., software or firmware to perform welding processes), and/or any other appropriate data. Examples of stored data for a welding application include an attitude (e.g., orientation) of a welding torch, a distance between the contact tip and a workpiece, a voltage, a current, welding device settings, and so forth.

(126) The memory device **1824** may include a volatile memory, such as random access memory (RAM), and/or a nonvolatile memory, such as read-only memory (ROM). The memory device **1824** and/or the storage device(s) **1823** may store a variety of information and may be used for various purposes. For example, the memory device **1824** and/or the storage device(s) **1823** may store processor executable instructions **1825** (e.g., firmware or software) for the processor **1820** to execute. In addition, one or more control regimes for various welding processes, along with associated settings and parameters, may be stored in the storage device **1823** and/or memory device **1824**, along with code configured to provide a specific output (e.g., initiate wire feed, enable gas flow, capture welding current data, detect short circuit parameters, determine amount of spatter) during operation.

(127) In some examples, the welding power flows from the power converter **1810** through a weld cable **1826**. The example weld cable **1826** is attachable and detachable from weld studs at each of the power supply **1708, 1710** (e.g., to enable ease of replacement of the weld cable **1826** in case of wear or damage). Furthermore, in some examples, welding data is provided with the weld cable **1826** such that welding power and weld data are provided and transmitted together over the weld cable **1826**. The communications transceiver **1818** is communicatively coupled to the weld cable **1826** to communicate (e.g., send/receive) data over the weld cable **1826**. The communications transceiver **1818** may be implemented based on various types of power line communications methods and techniques. For example, the communications transceiver **1818** may utilize IEEE standard P1901.2 to provide data communications over the weld cable **1826**. In this manner, the weld cable **1826** may be utilized to provide welding power from the power supply **1708, 1710** to the wire feeder and the welding tool **108**. Additionally or alternatively, the weld cable **1826** may be used to transmit and/or receive data communications to/from the wire feeder and the welding tool **108**. The communications transceiver **1818** is communicatively coupled to the weld cable **1826**, for example, via cable data couplers **1827**, to characterize the weld cable **1826**, as described in more detail below. The cable data coupler **1827** may be, for example, a voltage or current sensor.

(128) In some examples, the power supply **1708, 1710** includes or is implemented in a wire feeder.

(129) The example communications transceiver **1818** includes a receiver circuit **1821** and a transmitter circuit **1822**. Generally, the receiver circuit **1821** receives data transmitted by the wire feeder via the weld cable **1826** and the transmitter circuit **1822** transmits data to the wire feeder via the weld cable **1826**. As described in more detail below, the communications transceiver **1818** enables remote configuration of the power supply **1708, 1710** from the location of the wire feeder and/or compensation of weld voltages by the power supply **1708, 1710** using weld voltage feedback information transmitted by the wire feeder **104**. In some examples, the receiver circuit **1821** receives communication(s) via the weld circuit while weld current is flowing through the weld circuit (e.g., during a welding-type operation) and/or after the weld current has stopped flowing through the weld circuit (e.g., after a welding-type operation). Examples of such communications include weld voltage feedback information measured at a device that is remote from the power supply **1708, 1710** (e.g., the wire feeder) while the weld current is flowing through the weld circuit

(130) Example implementations of the communications transceiver **1818** are described in U.S. Pat. No. 9,012,807. The entirety of U.S. Pat. No. 9,012,807 is incorporated herein by reference. However, other implementations of the communications transceiver **1818** may be used.

(131) The example wire feeder **104** also includes a communications transceiver **1819**, which may be similar or identical in construction and/or function as the communications transceiver **1818**.

(132) In some examples, a gas supply **1828** provides shielding gases, such as argon, helium, carbon dioxide, and so forth, depending upon the welding application. The shielding gas flows to a valve **1830**, which controls the flow of gas, and if desired, may be selected to allow for modulating or regulating the amount of gas supplied to a welding application. The valve **1830** may be opened, closed, or otherwise operated by the controller **1812** to enable, inhibit, or control gas flow (e.g., shielding gas) through the valve **1830**. Shielding gas exits the valve **1830** and flows through a cable **1832** (which in some implementations may be packaged with the welding power output) to the wire feeder which provides the shielding gas to the welding application. In some examples, the power supply **1708, 1710** does not include the gas supply **1828**, the valve **1830**, and/or the cable **1832**.

(133) FIG. **19** illustrates another example liquid-cooled welding torch **1900**. Like the torch **200** of FIGS. **2-14B**, the example liquid-cooled welding torch **1900** resistively preheats an electrode wire **114** via multiple contact points (e.g., contact tips) in the torch **1900**.

(134) The torch **1900** of FIG. **19** includes a welding assembly **1902**, a mounting assembly **1904**, a weld cable **1906**, liquid cooling assemblies **1908, 1910, 1912, 1914**, and a power and liquid transfer assembly **1916**. The example liquid-cooled welding torch **1900** may be used to replace

conventional robotic welding torches with resistive preheating-enabled welding torches having a same tool center point (TCP).

(135) The welding assembly **1902** receives weld current and preheating current, conducts the weld current to the electrode wire **114**, and conducts the preheating current through a portion of the electrode wire **114**. The example welding assembly **1902** is liquid-cooled by liquid provided via the liquid cooling assemblies **1908-214**. The example welding assembly **1902** of FIG. **19** receives the weld current via one or more of the weld cable **1906**, the liquid cooling assemblies **1908** and **1912**, and/or the liquid cooling assemblies **1910** and **1914**. Because the workpiece provides the return path for the weld current to the power supply, no return path is provided via the weld cable **1906** or the liquid cooling assemblies **1908**. The weld cable **1906** is a air-cooled (or gas-cooled) cable. However, the weld cable **1906** may also be liquid-cooled.

(136) The example welding assembly **1902** receives the preheating current via the weld cable **1906**, the liquid cooling assemblies **1908** and **1912**, and/or the liquid cooling assemblies **1910** and **1914**. In the example of FIG. **19**, the weld current is conducted via a different one of the weld cable **1906**, the liquid cooling assemblies **1908** and **1912**, or the liquid cooling assemblies **1910** and **1914** than the preheating current that has the same polarity (i.e., current flow direction). The welding assembly **1902** conducts the preheating current through a section of the electrode wire **114** to heat the electrode wire via resistive heating (e.g., I²R heating). The preheat current then returns to a preheating power supply via a different one of weld cable **1906**, the liquid cooling assemblies **1908** and **1912**, or the liquid cooling assemblies **1910** and **1914** to complete a preheating circuit.

(137) In the example of FIG. **19**, the weld current path, the preheating current supply path, and the preheating current return path may all be different ones of the weld cable **1906**, the liquid cooling assemblies **1908** and **1912**, and the liquid cooling assemblies **1910** and **1914**. In some examples, the weld current path may be superimposed with the preheating current supply path or the preheating current return path to reduce the net current in the conductor. For example, if the weld current is 300 A and the preheating current is 100 A, the weld current and the preheating current may be superimposed to result in a net current of 1900 A.

(138) As described in more detail below, the welding assembly **1902** and the liquid cooling assemblies **1912**, **1914** may be separated from the remainder of the liquid-cooled welding torch **1900** via the power and liquid transfer assembly **1916**, and may be simultaneously separated from the mounting assembly **1904**.

(139) FIG. **20** is an exploded view of the example welding assembly **1902** of the welding torch **1900** of FIG. **19**. The welding assembly **1902** includes a front portion **2002** and a rear portion **2004**. FIG. **21** is a cross-sectional plan view of the example welding assembly **1902** of FIG. **19** in which the front portion **2002** is coupled to the rear portion **2004**. As described in more detail below, the front portion **2002** is detachable from the rear portion to enable access to a rear contact tip **2006** (e.g., a preheating contact tip).

(140) The front portion **2002** includes a nozzle **2008**, a nozzle insulator **2010**, a front contact tip **2012**, a diffuser **2014**, a wire guide **2016**, a cooling body **2018**, a hand nut **2020**, and a hand nut insulator **2022**. The rear portion **2004** includes a valve assembly **2024**, a valve assembly insulator **2026**, and a valve assembly housing **2028**.

(141) The example hand nut **2020** secures the cooling body **2018**, and the components **2008-2016** connected to the cooling body **2018**, to the rear portion **2004**. In the example of FIG. **20**, the hand nut **2020** has internal screw threads to be threaded onto external screw threads **2030** of the valve assembly **2024**. A tapered edge **2032** of the hand nut **2020** mates with a shoulder of the cooling body **2018** to force the cooling body **2018** toward the valve assembly **2024**. The hand nut insulator **2022** electrically insulates the hand nut to reduce or prevent an operator contacting weld voltage and/or preheating voltage via the hand nut **2020**.

(142) The example valve assembly **2024** includes fluid valves **2034a**, **2034b** positioned within fluid channels **2036a**, **2036b**, respectively. The fluid channels **2036a**, **2036b** are in fluid communication

with the liquid cooling assemblies **1912**, **1914** to circulate fluid through the welding assembly **1902**. The example valves **2034a**, **2034b** are Shrader valves that cut off fluid flow when the valves are not actuated. To actuate the valves, the example cooling body **2018** includes valve actuators **2038a**, **2038b**, which are located within channels **2040a**, **2040b** of the cooling body **2018**. The valve actuators **2038a**, **2038b** actuate the valves **2034a**, **2034b** when the front portion **2002** (including the cooling body **2018**) is secured to the rear portion **2004**.

(143) When the valves are actuated, the cooling body **2018** is in fluid communication with the liquid cooling assemblies **1912**, **1914**. The example cooling body **2018** includes one or more internal channels to direct fluid from one of the valve actuators **2038a**, **2038b** to a second one of the valve actuators **2038a**, **2038b**. In other words, one of the valve actuators **2038a**, **2038b** is an inlet to the channel(s) **2102** in the cooling body **2018** from one of the liquid cooling assemblies **1912**, **1914** and the other of the valve actuators **2038a**, **2038b** is an outlet from the channel(s) **2102** to the other of the liquid cooling assemblies **1912**, **1914**. The example channels **2102** run circumferentially within the cooling body **2018** between the valve actuators **2038a**, **2038b** to transfer heat from the nozzle **2008** to the fluid within the channels **2102**.

(144) The example nozzle **2008** includes internal threads **2104** that couple the nozzle **2008** to a threaded ring **2042** coupled to an exterior of the cooling body **2018**. When coupled, heat from the nozzle **2008** is transferred to the cooling body **2018** for further dissipation to the cooling liquid.

(145) When secured (e.g., threaded together), the cooling body **2018** and the valve assembly **2024** are in electrical contact to transfer welding current and/or preheating current with the front contact tip **2012** via the diffuser **2014**. The welding current and/or preheating current are conducted via one or more of the liquid cooling assemblies **1912**, **1914**. For example, one or both of the liquid cooling assemblies **1912**, **1914** include a conductor layer electrically coupled to the front contact tip **2012** via the diffuser **2014**, the cooling body **2018**, and the valve assembly **2024**. Preheating current is conducted to the rear contact tip **2006** from the weld cable **1906** via a torch neck **2044**, which includes one or more layers of conductors, a wire liner to transfer the electrode wire **114**, and an annulus to provide gas flow to the weld assembly **1902** for output to the weld via the diffuser **2014**.

(146) As illustrated in FIG. **20**, the liquid cooling assemblies **1912**, **1914** may be secured to the torch neck **2044** via bracket **2046** or other attachment technique.

(147) The example wire guide **2016** may be similar or identical to the wire guide **308** of FIG. **3**. The wire guide **2016** is held within a bore in the diffuser **2014**. The nozzle insulator **2010** electrically and thermally insulates the nozzle **2008** from the front contact tip **2012** and the diffuser **2014**. The nozzle **2008** is electrically insulated from the cooling body **2018** by one or more additional electrical insulator(s) **2110** located on the cooling body **2018**. The example electrical insulator(s) **2110** may be thermally conductive to conduct heat from the nozzle **2008** to the cooling body **2018**. Polyether ether ketone (PEEK) and/or other thermoplastics may be used to implement the example electrical insulator(s) **2110**.

(148) The example cooling body **2018** may include any number of electrical insulators and/or fluid seals to enable conduction of current to the diffuser **2014**, reduce or prevent conduction of current to unneeded or undesired components (e.g., exterior components), and/or to reduce or prevent fluid leaks from the cooling body **2018**. In the example of FIG. **21**, the cooling body **2018** includes an inner body **2106** defining the channel(s) **2102** and a cover **2108** configured to enclose the channels **2102**.

(149) The example rear contact tip **2006** may be similar or identical to the second contact tip **314** of FIGS. **3** and **13A-13C**.

(150) FIG. **22** is a cross-sectional plan view of the example power and liquid transfer assembly **1916** of FIG. **19**. FIG. **23** is a partially exploded elevation view of the example power and liquid transfer assembly **1916** of FIG. **19**.

(151) The example power and liquid transfer assembly **1916** is similar to the power and liquid transfer assembly **216** of FIGS. **2**, **3**, and **8**, in that the power and liquid transfer assembly **1916**

includes multiple liquid transfer valves, corresponding valve actuators, and one or more electrical power transfer sockets, and in that the power and liquid transfer assembly **1916** enables disconnection of the liquid cooling assemblies **1912**, **1914** from the power and liquid transfer assembly **1916** and cuts off fluid transfer in response to disconnection.

(152) The example power and liquid transfer assembly **1916** is coupled to the liquid cooling assemblies **1908**, **1910**, which provide supply and return lines for the cooling liquid to a liquid cooler. The example liquid cooling assembly **1908** includes an internal conductor **2202** to conduct welding current (e.g., to the welding power supply **1710** of FIG. 17) and/or preheating current (e.g., to the preheating power supply **1708**). The conductor **2202** is in electrical contact with a socket **2204**, which permits fluid to flow from an exterior of the socket **2204** to an interior of the socket **2204**, into which a fluid fitting **2206** of the power and liquid transfer assembly **1916** is fitted to make a fluid connection with the liquid cooling assembly **1908** and electrical contact with the socket **2204**. In the example of FIG. 22, the liquid cooling assembly **1910** only includes tubing to transfer liquid, and does not include an internal conductor or a socket. In other examples, the liquid cooling assembly **1910** also includes a conductor and has a construction similar or identical to the liquid cooling assembly **1908**.

(153) Each of the example liquid cooling channels of the power and liquid transfer assembly **1916** includes a liquid shutoff valve **2208a**, **2208b** within a fluid socket **2209a**, **2209b**. The channel coupled to the liquid cooling assembly **1908** (e.g., carrying current and liquid) includes a power transfer socket **2210**. The channel coupled to the liquid cooling assembly **1910** (e.g., carrying only liquid) may include a non-conductive socket **2212** having similar, the same, or different dimensions as the power transfer socket **2210**.

(154) The power cable socket **2210** receives a power connector pin **2214** of the liquid cooling assembly **1912** to transfer cooling liquid and welding current and/or preheating current to the liquid cooling assembly **1912**. The nonconductive socket **2212** likewise receives a power connector pin or other pin corresponding to the dimensions of the socket **2212**. The power transfer socket **2210** enables insertion of the power connector pin **2214**, and transfers current to and/or from an inserted power connector pin **2214**. An example power transfer socket that may be used to implement the power transfer socket **2210** is a PowerBud® power contact, sold by Methode Electronics, Inc., which provides multiple contact points between the power transfer socket and an inserted power connector pin **2214**.

(155) The liquid shutoff valves **2208a**, **2208b** selectively permit flow of liquid from liquid cooling assemblies **1908**, **1910** to the sockets **2210**, **2212** and to a connected liquid cooling assembly **1912**, **1914**. The example liquid shutoff valves **2208a**, **2208b** are Schrader valves. However, other types of valves may be used to implement the liquid shutoff valves **2208a**, **2208b**. When a power connector pin **2214**, **2216** is inserted (e.g., fully inserted) into the sockets **2210**, **2212**, the power connector pin **2214**, **2216** displaces (e.g., unseats) a stem **2218** from a core **2220** of the valve **2208a**, **2208b**, which permits liquid to flow to and/or from the hose liquid cooling assemblies **1908-1914**. When the power connector pins **2214**, **2216** are removed or partially removed, the stems **2218** are forced back into the cores **2220** and stop flow of liquid.

(156) A hose **2222** of the liquid cooling assemblies **1908** is coupled to the fluid socket **2209a** via a ferrule **2224**. The example the example sockets **2209a**, **2209b** and/or the hoses **2222** include hose barbs to secure the hoses **2222**. However, other methods of securing the hose to the sockets **2209a**, **2209b** may be used, such as clamps, compression fittings, or any other hose fittings.

(157) The example power and liquid transfer assembly **1916** may operate as described above with reference to the power and liquid transfer assembly **216** of FIG. 8.

(158) As illustrated in FIG. 23, the liquid cooling assemblies **1912**, **1914** may be secured to the power and liquid transfer assembly **1916** at least partly using a cover **2302** configured to prevent disconnection of the power connector pins **2214**, **2216** from the power and liquid transfer assembly **1916**. For example, the cover **2302** may include a shoulder and/or other features configured to

prevent movement of the power connector pins **2214**, **2216** away from the power and liquid transfer assembly **1916**. The cover **2302** may be secured by a fastener, such as a bolt, and/or any other type of fastener or fastening technique.

(159) While the present method and/or system has been described with reference to certain implementations, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present method and/or system. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. For example, systems, blocks, and/or other components of disclosed examples may be combined, divided, re-arranged, and/or otherwise modified. Therefore, the present method and/or system are not limited to the particular implementations disclosed. Instead, the present method and/or system will include all implementations falling within the scope of the appended claims, both literally and under the doctrine of equivalents.

(160) All documents cited herein, including journal articles or abstracts, published or corresponding U.S. or foreign patent applications, issued or foreign patents, or any other documents are each entirely incorporated by reference herein, including all data, tables, figures, and text presented in the cited documents.

Claims

1. A welding torch, comprising: a first contact tip configured to conduct welding current to a consumable electrode; a second contact tip configured to conduct preheating current to the consumable electrode; a cooling assembly configured to transfer heat from at least the first contact tip to a coolant, the cooling assembly comprising an electrically conductive cooling body that defines a flow path for the coolant, wherein the cooling body conducts the welding current; wherein the first contact tip and the cooling assembly are removable from the welding torch as a single unit.
2. The welding torch as defined in claim 1, wherein the second contact tip is removable from the welding torch with the cooling assembly and the first contact tip as part of the single unit.
3. The welding torch as defined in claim 1, further comprising a torch neck configured to couple the first contact tip, the second contact tip, and the cooling assembly to a torch body, the torch neck being removable from the welding torch with the cooling assembly and the first contact tip as part of the single unit.
4. The welding torch as defined in claim 1, further comprising fluid lines configured to direct fluid to the cooling assembly and away from the cooling assembly, the fluid lines being removable from the welding torch with the cooling assembly and the first contact tip as part of the single unit.
5. The welding torch as defined in claim 4, further comprising a liquid transfer assembly configured to transfer fluid to and from the fluid lines, the liquid transfer assembly comprising valves configured to seal fluid into the liquid transfer assembly when the fluid lines are removed.
6. The welding torch as defined in claim 1, further comprising a gas diffuser configured to electrically and thermally couple the first contact tip to the cooling assembly, the gas diffuser being removable from the welding torch as part of the single unit.
7. The welding torch as defined in claim 6, the diffuser comprising: a first exterior portion configured to be coupled to the cooling assembly; a first interior portion configured to receive welding gas from an interior of the cooling assembly; a second interior portion configured to receive the first contact tip and to conduct current to the first contact tip; and a gas diffusion port configured to conduct welding gas from the first interior portion to a second exterior portion, the gas diffusion port configured to conduct the welding gas in a radial direction of the diffuser.
8. The welding torch as defined in claim 7, further comprising: an insulator; and a nozzle coupled to an exterior of the gas diffuser via the insulator, the insulator and the nozzle being removable

from the welding torch as part of the single unit.

9. The welding torch as defined in claim 1, wherein the cooling body comprises: a liquid coolant inlet port; a liquid coolant outlet port; and a channel to channel the liquid coolant from the liquid coolant inlet port to the liquid coolant outlet port, the channel providing the flow path for the liquid coolant.

10. The welding torch as defined in claim 9, further comprising a valve assembly configured to be mechanically and fluidly coupled to the cooling body.

11. The welding torch as defined in claim 10, wherein the valve assembly is removable from the welding torch as part of the single unit.

12. The welding torch as defined in claim 9, wherein the cooling body comprises a cover to enclose the channel.
